

The Normal Form Lemma for Smooth h -Cobordisms

1 Goal of the lecture

Let $(W; M_0, M_1)$ be a compact smooth cobordism of dimension n . Thus

$$\partial W = M_0 \sqcup M_1,$$

where M_0 is regarded as the incoming boundary and M_1 as the outgoing boundary. We say that W is an h -cobordism if both inclusions

$$M_0 \hookrightarrow W, \quad M_1 \hookrightarrow W$$

are homotopy equivalences.

The purpose of this lecture is to explain why, in dimension at least six, every smooth h -cobordism can be put into a particularly simple handle-theoretic form.

Theorem 1.1 (Normal form lemma). *Let $(W; M_0, M_1)$ be a compact smooth h -cobordism of dimension $n \geq 6$. Let q be an integer satisfying*

$$2 \leq q \leq n - 3.$$

Then W admits a handlebody decomposition relative to M_0 using only handles of index q and $q + 1$. Equivalently, there is a diffeomorphism relative to M_0

$$W \cong M_0 \times [0, 1] \cup \bigcup_{i=1}^{p_q} h_i^q \cup \bigcup_{j=1}^{p_{q+1}} h_j^{q+1},$$

where h_i^r denotes a handle of index r .

This result is often called the normal form lemma. It is the geometric step that prepares the h -cobordism for the algebraic part of the s -cobordism theorem. Once the handle decomposition has only two adjacent dimensions, the relative chain complex

$$C_*(\widetilde{W}, \widetilde{M}_0)$$

has only one possibly nonzero differential. That differential is represented, after choosing lifts of the handles, by a matrix over the group ring $\mathbb{Z}[\pi]$, where

$$\pi := \pi_1(W) \cong \pi_1(M_0).$$

In the h -cobordism case this two-term complex is acyclic, so the matrix is invertible in the corresponding algebraic sense. This is the point at which Whitehead torsion enters the proof of the s -cobordism theorem.

The guiding slogan for the proof is:

A low-index handle can be removed at the cost of creating a higher-index handle.

More precisely, under suitable hypotheses, a q -handle can be eliminated by introducing a new $(q + 2)$ -handle. Repeating this operation pushes handles upward until no handles remain below the chosen index q . Then we reverse the cobordism and apply the same argument to the dual handle decomposition. In the dual picture, high-index handles in the original decomposition become low-index handles, so the same method removes them.

The argument has three main ingredients: handle cancellation, the acyclicity of the relative chain complex, and the Whitney trick. The dimension hypotheses

$$n \geq 6, \quad 2 \leq q \leq n - 3$$

are used to ensure that the Whitney trick and the required embedded and framed constructions can be carried out in the boundary levels of the cobordism.

These notes follow the standard handle-theoretic route. For additional background on handle attachments, handle decompositions, dual handles, handle cancellation, the Whitney trick, and the proof of the h -cobordism theorem, see also Friedl's lecture notes [Fri], especially the chapters on handle structures, Morse theory, and the h -cobordism theorem.

Example 1.2 (What the theorem says in dimension six). If $n = 6$, then the only possible value of q is $q = 2$. The normal form lemma therefore says that every smooth six-dimensional h -cobordism admits a handle decomposition relative to M_0 using only 2-handles and 3-handles. This is the first nontrivial case and is a good model to keep in mind while reading the proof.

Example 1.3 (The trivial product cobordism). The product cobordism

$$W = M_0 \times [0, 1]$$

is an h -cobordism in the most elementary way: both boundary inclusions are homotopy equivalences. Its normal form has no handles at all. Equivalently, for any allowed q , one may take

$$p_q = p_{q+1} = 0.$$

This example is useful because it reminds us that the phrase “only q - and $(q + 1)$ -handles” allows the possibility that no such handles occur.

Example 1.4 (Allowed values of q). The inequality $2 \leq q \leq n - 3$ says that the two remaining handle indices must stay away from the bottom and top of the decomposition. For small dimensions the possibilities are:

n	allowed q	possible normal-form indices
6	2	(2, 3)
7	2, 3, 4	(2, 3), (3, 4), (4, 5)
8	2, 3, 4, 5	(2, 3), (3, 4), (4, 5), (5, 6).

The proof works uniformly in all these cases, but the Whitney trick is the geometric reason for excluding $q = 1$ and $q = n - 2$.

2 Handlebody decompositions relative to one boundary component

An r -handle in an n -dimensional cobordism is

$$h^r = D^r \times D^{n-r}.$$

It is attached along

$$S^{r-1} \times D^{n-r} \subset \partial(D^r \times D^{n-r}).$$

The disk

$$D^r \times \{0\}$$

is called the core of the handle, and the disk

$$\{0\} \times D^{n-r}$$

is called the cocore. The boundary of the cocore,

$$\{0\} \times S^{n-r-1},$$

is called the belt sphere, or transverse sphere, of the handle.

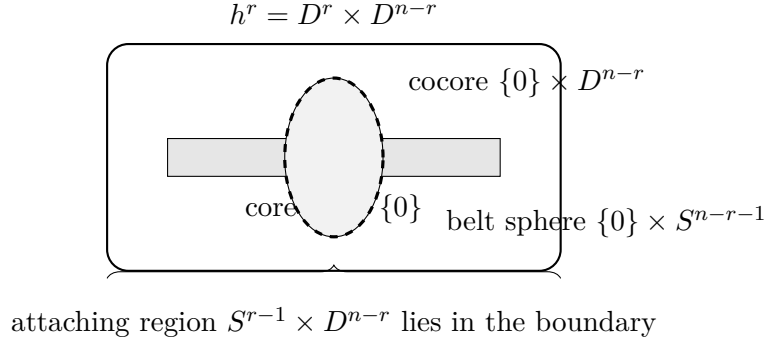


Figure 1: A schematic picture of an r -handle. The picture is not dimensionally faithful; it only records the roles of the core, cocore, and belt sphere.

A handlebody decomposition of W relative to M_0 is a presentation

$$W \cong M_0 \times [0, 1] \cup \bigcup_{i=1}^{p_0} h_i^0 \cup \bigcup_{i=1}^{p_1} h_i^1 \cup \cdots \cup \bigcup_{i=1}^{p_n} h_i^n,$$

where the decomposition is relative to the incoming boundary

$$M_0 = M_0 \times \{0\}.$$

We write W_r for the partial cobordism obtained after all handles of index at most r have been attached:

$$W_r := M_0 \times [0, 1] \cup \bigcup_{i=1}^{p_0} h_i^0 \cup \cdots \cup \bigcup_{i=1}^{p_r} h_i^r.$$

The boundary component that changes as handles are attached will be denoted by

$$\partial_1 W_r.$$

Thus $\partial_1 W_r$ is the part of ∂W_r not equal to the old boundary $M_0 \times \{0\}$.

The geometric content of the normal form lemma is that a complicated handle decomposition can be simplified until only a narrow band of indices remains. However, handles cannot usually be deleted one at a time. The basic mechanism is instead cancellation of pairs of consecutive indices.

3 The cancellation lemma

The fundamental cancellation criterion is the following.

Lemma 3.1 (Handle cancellation). *Let V be an n -dimensional compact cobordism. Suppose a q -handle h^q is attached to V , and then a $(q+1)$ -handle h^{q+1} is attached. If the attaching sphere*

$$S^q \times \{0\}$$

of h^{q+1} meets the belt sphere of h^q transversely in exactly one point, then the two handles cancel. In other words,

$$V \cup h^q \cup h^{q+1} \cong V$$

relative to the incoming boundary.

The condition of a single transverse intersection is the smooth analogue of an elementary collapse in CW theory. The q -handle and the $(q+1)$ -handle are attached in the simplest possible nontrivial way, and together they do not change the diffeomorphism type of the cobordism relative to the incoming boundary.

In a handlebody decomposition with many q -handles, the attaching sphere of a new $(q+1)$ -handle must meet the belt sphere of the particular q -handle we want to cancel exactly once and must be disjoint from the belt spheres of all other q -handles. Much of the proof is devoted to arranging precisely this situation.

Remark 3.2. The cancellation lemma is local once its hypothesis has been achieved. Arranging that hypothesis is global. The attaching sphere may wind around the cobordism, and its algebraic intersections with belt spheres are measured in the group ring $\mathbb{Z}[\pi]$, where $\pi = \pi_1(W)$. This is why the proof needs both geometry and algebra.

Example 3.3 (The basic cancellation picture). The easiest example is a single 0-handle together with a 1-handle whose attaching sphere meets the belt sphere of the 0-handle once. Their union is diffeomorphic to a collar, so the pair cancels. Higher-index cancellation is the same phenomenon in higher dimensions: a $(q+1)$ -handle cancels a q -handle when its attaching sphere passes exactly once through the relevant belt sphere.

4 The elimination lemma: trading a handle upward

The next tool explains how a q -handle can be eliminated at the cost of introducing a new $(q+2)$ -handle. This is the central geometric operation in the proof.

Assume that the handlebody decomposition has no handles of index below q :

$$W \cong M_0 \times [0, 1] \cup \bigcup_{i=1}^{p_q} h_i^q \cup \bigcup_{j=1}^{p_{q+1}} h_j^{q+1} \cup \bigcup_{k=1}^{p_{q+2}} h_k^{q+2} \cup \dots$$

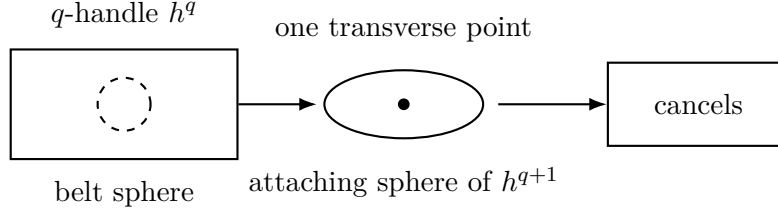


Figure 2: Schematic cancellation: once the attaching sphere of the $(q + 1)$ -handle meets the belt sphere of the q -handle transversely in a single point, the two handles cancel.

Choose one q -handle, say $h_{i_0}^q$. Suppose we can find a framed embedding

$$\varphi : S^q \times D^{n-q-1} \longrightarrow \partial_1 W_q$$

with the following two properties.

First, the core sphere

$$\varphi(S^q \times \{0\})$$

is isotopic in $\partial_1 W_q$ to an embedding that meets the belt sphere of $h_{i_0}^q$ transversely in exactly one point and is disjoint from the belt spheres of all other q -handles.

Second, after the existing $(q + 1)$ -handles have been attached, the same core sphere becomes isotopic to a trivial framed embedding in $\partial_1 W_{q+1}$.

Here “trivial” means that the framed sphere is the standard attaching sphere of a cancelling $(q + 1)$ -handle; equivalently, it bounds a suitably framed embedded disk after passing to the boundary level where the $(q + 1)$ -handles have been attached.

Lemma 4.1 (Elimination lemma). *Under the assumptions above, W is diffeomorphic relative to M_0 to a handlebody decomposition in which the handle $h_{i_0}^q$ has been removed and one new $(q + 2)$ -handle has been added. All other handles are unchanged up to isotopy and handle slides.*

Idea of the proof. Because the framed sphere becomes trivial after the existing $(q + 1)$ -handles are attached, we may introduce a cancelling pair consisting of a new $(q + 1)$ -handle and a new $(q + 2)$ -handle. Adding such a cancelling pair does not change the diffeomorphism type of the cobordism.

Now use the first property of φ . The attaching sphere of the newly created $(q + 1)$ -handle can be arranged to meet the belt sphere of the chosen q -handle $h_{i_0}^q$ exactly once and to miss the belt spheres of all other q -handles. By Lemma 3.1, this new $(q + 1)$ -handle cancels $h_{i_0}^q$. After this cancellation, the only remaining trace of the operation is the newly created $(q + 2)$ -handle.

Thus one has traded one q -handle for one $(q + 2)$ -handle. Geometrically, we first create a harmless cancelling pair. Then we use the lower handle $h_{i_0}^q$ to cancel the new $(q + 1)$ -handle instead of allowing that new $(q + 1)$ -handle to cancel the new $(q + 2)$ -handle. The original q -handle disappears, and the new $(q + 2)$ -handle remains.

Example 4.2 (Index bookkeeping in a handle trade). In an 8-dimensional cobordism, eliminating a 2-handle by the elimination lemma creates a new 4-handle. Thus this operation does not necessarily reduce the total number of handles. Its purpose is different: it raises the lowest index present in the decomposition. If the goal is to reach $q = 3$, trading away all 2-handles is exactly what is needed.

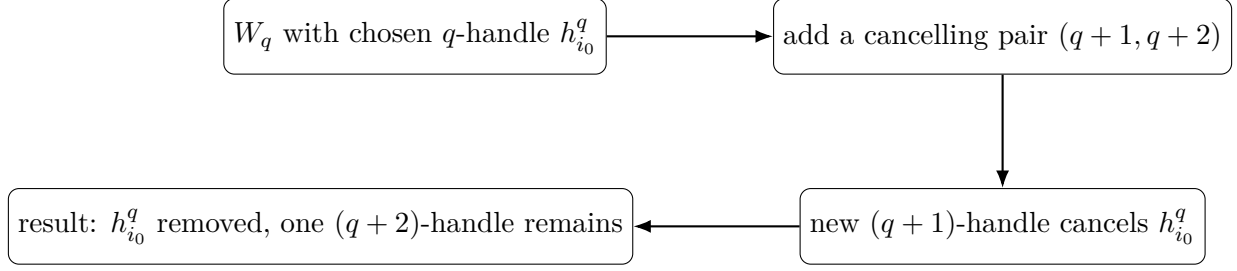


Figure 3: The geometric idea behind the elimination lemma: create a cancelling pair, then “use up” the new $(q+1)$ -handle to cancel the chosen q -handle, leaving a new $(q+2)$ -handle behind.

The elimination lemma reduces the main proof to a construction problem. Given a chosen q -handle, we must construct an attaching sphere satisfying the two properties above. The construction uses three ingredients:

1. the acyclicity of the relative chain complex $C_*(\widetilde{W}, \widetilde{M}_0)$;
2. the ability to modify an attaching sphere by adding boundaries of existing $(q+1)$ -handles;
3. the Whitney trick, which removes algebraically cancelling pairs of intersection points.

5 Removing 0- and 1-handles

Before the main induction begins, we remove handles of index 0 and 1. For an h -cobordism, the inclusion

$$M_0 \hookrightarrow W$$

is a homotopy equivalence. In particular, it is 1-connected in the following sense: it is bijective on path components and surjective on fundamental groups.

Lemma 5.1 (No 0- and 1-handles). *Let W be an n -dimensional compact smooth cobordism with $n \geq 6$. If the inclusion*

$$M_0 \hookrightarrow W$$

is 1-connected, then W admits a handlebody decomposition relative to M_0 with no handles of index 0 or 1.

Sketch of proof. First consider a 0-handle. Since $M_0 \rightarrow W$ is bijective on path components, the component created by this 0-handle must be connected to the old boundary by some 1-handle. After arranging the ordering of handles and, if necessary, performing standard handle slides, the core of such a 1-handle gives the required single transverse intersection with the belt sphere of the 0-handle. Lemma 3.1 then removes the 0-handle together with that 1-handle.

Now consider a 1-handle. To cancel it, we need a suitable 2-handle whose attaching circle meets the belt sphere of the chosen 1-handle once. The surjectivity of

$$\pi_1(M_0) \longrightarrow \pi_1(W)$$

allows a path running through the 1-handle to be closed up by a path near the old boundary so that the resulting loop is null-homotopic after the appropriate higher handles are attached. Since $n \geq 6$,

a null-homotopy disk can be arranged, by general position, to be embedded and to have interior disjoint from the relevant lower-dimensional data. After choosing the required framing, this loop thickens to the attaching region of a 2-handle. The elimination lemma then removes the chosen 1-handle, possibly creating a 3-handle.

Repeating this process removes all 0- and 1-handles.

For an h -cobordism, the hypotheses of Lemma 5.1 are automatically satisfied. Therefore, from this point on, we may assume that the handlebody decomposition begins in index 2:

$$W \cong M_0 \times [0, 1] \cup \bigcup_i h_i^2 \cup \bigcup_i h_i^3 \cup \dots.$$

6 The relative chain complex from handles

Let

$$\pi := \pi_1(W) \cong \pi_1(M_0),$$

where the isomorphism is induced by the inclusion $M_0 \hookrightarrow W$. Choose a universal cover

$$\widetilde{W} \longrightarrow W,$$

and let $\widetilde{M}_0$ denote the full preimage of M_0 in $\widetilde{W}$. We use the convention that cellular chains on the universal cover are free left $\mathbb{Z}[\pi]$ -modules.

The handle decomposition of W relative to M_0 lifts to a relative cellular decomposition of the pair

$$(\widetilde{W}, \widetilde{M}_0).$$

Each r -handle of W , after choosing one lift, gives a basis element of the free $\mathbb{Z}[\pi]$ -module

$$C_r(\widetilde{W}, \widetilde{M}_0).$$

We denote the basis element corresponding to a chosen lift of h_i^r by

$$[\tilde{h}_i^r] \in C_r(\widetilde{W}, \widetilde{M}_0).$$

Thus, if there are p_r handles of index r , then

$$C_r(\widetilde{W}, \widetilde{M}_0) \cong \mathbb{Z}[\pi]^{p_r}.$$

The boundary homomorphisms will be denoted by

$$d_r : C_r(\widetilde{W}, \widetilde{M}_0) \longrightarrow C_{r-1}(\widetilde{W}, \widetilde{M}_0).$$

Since $M_0 \hookrightarrow W$ is a homotopy equivalence, the pair (W, M_0) is homotopy equivalent to a point relative to M_0 . Equivalently, the lifted relative chain complex is acyclic:

$$H_*(\widetilde{W}, \widetilde{M}_0) = 0.$$

This acyclicity is the algebraic reason why lower handles can be eliminated.

Suppose, for example, that there are no handles below index q . Then

$$C_{q-1}(\widetilde{W}, \widetilde{M}_0) = 0.$$

Hence

$$d_q : C_q(\widetilde{W}, \widetilde{M}_0) \longrightarrow C_{q-1}(\widetilde{W}, \widetilde{M}_0)$$

is the zero map. Since the homology in degree q vanishes, we have

$$0 = H_q(\widetilde{W}, \widetilde{M}_0) = \frac{\ker d_q}{\operatorname{im} d_{q+1}}.$$

But $\ker d_q = C_q(\widetilde{W}, \widetilde{M}_0)$, because $d_q = 0$. Therefore

$$\operatorname{im} d_{q+1} = C_q(\widetilde{W}, \widetilde{M}_0).$$

Thus

$$d_{q+1} : C_{q+1}(\widetilde{W}, \widetilde{M}_0) \longrightarrow C_q(\widetilde{W}, \widetilde{M}_0)$$

is surjective.

Consequently, for any chosen q -handle $h_{i_0}^q$, its basis vector can be written as

$$[\tilde{h}_{i_0}^q] = \sum_{j=1}^{p_{q+1}} x_j d_{q+1}([\tilde{h}_j^{q+1}])$$

for suitable coefficients

$$x_j \in \mathbb{Z}[\pi].$$

This equality is the central algebraic input of the proof. It says that every q -handle is algebraically the boundary of a formal $\mathbb{Z}[\pi]$ -linear combination of $(q+1)$ -handles. The modification lemma will realize this formal combination geometrically.

Example 6.1 (A toy two-term chain complex). Suppose, after simplifying the decomposition, that there is one q -handle and one $(q+1)$ -handle. Then

$$C_{q+1}(\widetilde{W}, \widetilde{M}_0) \cong \mathbb{Z}[\pi], \quad C_q(\widetilde{W}, \widetilde{M}_0) \cong \mathbb{Z}[\pi].$$

The differential is multiplication by some element $a \in \mathbb{Z}[\pi]$:

$$d_{q+1}(1) = a.$$

If the relative complex is acyclic, then this map is an isomorphism, so a must be invertible in the appropriate sense. This simple model is the algebraic shadow of the final normal form.

Example 6.2 (The group ring when $\pi \cong \mathbb{Z}$). If $\pi \cong \mathbb{Z}$ with generator t , then

$$\mathbb{Z}[\pi] \cong \mathbb{Z}[t, t^{-1}].$$

A coefficient such as

$$2 - t + t^{-3}$$

means that the corresponding attaching sphere intersects chosen lifts of belt spheres in several translates: two positive contributions in one translate, one negative contribution in the translate indexed by t , and one positive contribution in the translate indexed by t^{-3} . The group-ring coefficient is therefore a compact algebraic record of how the attaching sphere winds around the fundamental group.

$$0 \longrightarrow C_{q+1}(\widetilde{W}, \widetilde{M}_0) \xrightarrow{d_{q+1}} C_q(\widetilde{W}, \widetilde{M}_0) \longrightarrow 0$$

After the normal form is reached, the relative chain complex is concentrated in two adjacent degrees.

Figure 4: The two-term relative chain complex that remains once only q - and $(q+1)$ -handles are present.

7 The homology lemma and the role of the Whitney trick

Let

$$2 \leq q \leq n-3.$$

Consider an embedding

$$f : S^q \hookrightarrow \partial_1 W_q.$$

Assume that f is transverse to the belt spheres of the q -handles. If B_i denotes the belt sphere of h_i^q , then

$$\dim S^q + \dim B_i = q + (n - q - 1) = n - 1 = \dim(\partial_1 W_q).$$

Thus the intersections

$$f(S^q) \cap B_i$$

are isolated points.

These intersections define an element of the relative chain group

$$C_q(\widetilde{W}, \widetilde{M}_0).$$

To see this, choose a lift $\tilde{f} : S^q \rightarrow \partial_1 \widetilde{W}_q$, and choose a lift $\tilde{B}_i$ of each belt sphere B_i . Every intersection point of $\tilde{f}(S^q)$ with a translate $g\tilde{B}_i$, $g \in \pi$, contributes

$$\pm g[\tilde{h}_i^q]$$

to $C_q(\widetilde{W}, \widetilde{M}_0)$. The sign is the local intersection number, and the group element g records which translate of the chosen lift of the belt sphere is met.

Thus we obtain an algebraic intersection chain

$$[\tilde{f}] = \sum_{i=1}^{p_q} \sum_{g \in \pi} \epsilon_{i,g} g[\tilde{h}_i^q] \in C_q(\widetilde{W}, \widetilde{M}_0),$$

where only finitely many coefficients $\epsilon_{i,g} \in \mathbb{Z}$ are nonzero.

The key geometric statement is the following.

Lemma 7.1 (Homology lemma). *Let $n \geq 6$ and $2 \leq q \leq n-3$. Let*

$$f : S^q \hookrightarrow \partial_1 W_q$$

be an embedding transverse to the belt spheres of the q -handles. Then f is isotopic in $\partial_1 W_q$ to an embedding that meets the belt sphere of $h_{i_0}^q$ transversely in exactly one point and is disjoint from the belt spheres of all other q -handles if and only if, after choosing a lift,

$$[\tilde{f}] = \pm g[\tilde{h}_{i_0}^q]$$

for some $g \in \pi$.

Idea of the proof. The implication from geometry to algebra is immediate. If $f(S^q)$ has exactly one transverse intersection with the belt sphere of $h_{i_0}^q$ and no intersections with the other belt spheres, then the lifted intersection chain consists of a single signed translate:

$$[\tilde{f}] = \pm g[\tilde{h}_{i_0}^q].$$

Conversely, suppose

$$[\tilde{f}] = \pm g[\tilde{h}_{i_0}^q].$$

After putting f in general position with respect to the belt spheres, this equality says that all additional intersection points cancel algebraically in pairs. More explicitly, every unwanted pair consists of two intersection points with opposite signs and the same group element in π . The equality of the group elements means that the loop formed by joining the two points along $f(S^q)$ and along the relevant belt sphere is null-homotopic in the appropriate complement. This is precisely the condition needed to construct a Whitney disk for that pair.

The high-dimensional Whitney trick then removes each algebraically cancelling pair by an isotopy of f . In the present range, the ambient boundary satisfies

$$\dim(\partial_1 W_q) = n - 1 \geq 5,$$

and the inequalities

$$q \geq 2, \quad n - q - 1 \geq 2$$

put both the sphere $f(S^q)$ and the belt spheres in the standard codimension range for the embedded and framed Whitney moves used in handle theory. The Whitney disks may therefore be chosen with the required interior-disjointness and framing, after the usual perturbations.

Repeating the Whitney move for all cancelling pairs leaves exactly one intersection with the belt sphere of $h_{i_0}^q$ and no intersections with the belt spheres of the other q -handles.

This is the main high-dimensional geometric input in the argument. Algebra tells us which pairs of intersection points should cancel; the Whitney trick performs the geometric cancellation.

Example 7.2 (A simple intersection chain). Suppose $q = 2$ and the lifted sphere $\tilde{f}(S^2)$ meets two translates of the belt sphere of h_1^2 : once positively in the translate indexed by the identity element and once negatively in the translate indexed by the same identity element. Then these two intersections contribute

$$[\tilde{h}_1^2] - [\tilde{h}_1^2] = 0.$$

The Whitney trick removes this cancelling pair geometrically. By contrast, if the sphere meets the belt sphere once positively and has no other intersections, then the chain is exactly $[\tilde{h}_1^2]$, which is the desired situation for cancellation.

Example 7.3 (Why the group element matters). Two unwanted intersection points with opposite signs do not automatically form a Whitney pair. In the universal cover, they must also occur in the same translate of the relevant belt sphere. In chain language, the cancellation

$$g[\tilde{h}_i^q] - g[\tilde{h}_i^q] = 0$$

corresponds to a geometric Whitney disk. By contrast,

$$g[\tilde{h}_i^q] - g'[\tilde{h}_i^q]$$

with $g \neq g'$ records two intersections that may have opposite local signs but lie in different fundamental-group positions. The equality in $\mathbb{Z}[\pi]$, not merely the equality of ordinary integer intersection numbers, is what detects the Whitney move needed here.

8 The modification lemma

We now need to construct an attaching sphere whose algebraic intersection chain is exactly the desired basis vector

$$[\tilde{h}_{i_0}^q].$$

The modification lemma provides this construction.

Start with a trivial framed embedding

$$\alpha : S^q \times D^{n-q-1} \hookrightarrow \partial_1 W_q.$$

Let the same symbol α denote its core sphere

$$\alpha : S^q \hookrightarrow \partial_1 W_q.$$

Since α is trivial, its algebraic intersection chain is zero:

$$[\tilde{\alpha}] = 0.$$

The idea is to modify α by connected summing it with attaching spheres of the existing $(q+1)$ -handles. This changes the algebraic intersection chain by adding boundaries of those $(q+1)$ -handles. At the same time, after the $(q+1)$ -handles have been attached, the modification becomes isotopic to the original embedding, because each attaching sphere bounds the core disk of its $(q+1)$ -handle.

Lemma 8.1 (Modification lemma). *Let*

$$f : S^q \hookrightarrow \partial_1 W_q$$

be a framed embedding. For any coefficients

$$x_1, \dots, x_{p_{q+1}} \in \mathbb{Z}[\pi],$$

there is a framed embedding

$$g : S^q \hookrightarrow \partial_1 W_q$$

such that:

- (i) g is isotopic to f inside $\partial_1 W_{q+1}$;
- (ii) after choosing compatible lifts,

$$[\tilde{g}] = [\tilde{f}] + \sum_{j=1}^{p_{q+1}} x_j d_{q+1}([\tilde{h}_j^{q+1}]).$$

Idea of the proof. It is enough to add one term at a time. Take the attaching sphere of a $(q+1)$ -handle h_j^{q+1} , and form a connected sum of $f(S^q)$ with that attaching sphere along an embedded tube in $\partial_1 W_q$. The resulting sphere is still embedded, after choosing the tube in general position.

Once the handle h_j^{q+1} has been attached, its attaching sphere bounds the core disk of that handle. Therefore the connected-sum operation can be undone by an isotopy inside $\partial_1 W_{q+1}$. This proves that the modified embedding is isotopic to the original one after the $(q+1)$ -handles have been attached.

On the chain level, connected summing with the attaching sphere of h_j^{q+1} adds the boundary

$$d_{q+1}([\tilde{h}_j^{q+1}])$$

to the algebraic intersection chain. A coefficient in $\mathbb{Z}[\pi]$ is a finite sum of signed group elements. By choosing the tube to wind around a loop representing a prescribed element $g \in \pi$, and by choosing the orientation of the connected-sum operation, one obtains a prescribed term

$$\pm g d_{q+1}([\tilde{h}_j^{q+1}]).$$

Repeating the construction for all terms appearing in the coefficients x_j gives

$$[\tilde{g}] = [\tilde{f}] + \sum_{j=1}^{p_{q+1}} x_j d_{q+1}([\tilde{h}_j^{q+1}]).$$

The modification lemma is the mechanism that translates the algebraic equality from the chain complex into a geometric attaching sphere.

Example 8.2 (How the modification lemma is used). Take $f = \alpha$ with $[\tilde{\alpha}] = 0$. If the surjectivity of d_{q+1} gives

$$[\tilde{h}_{i_0}^q] = d_{q+1}([\tilde{h}_1^{q+1}]) + g d_{q+1}([\tilde{h}_2^{q+1}]),$$

then the modification lemma builds a new sphere g whose algebraic intersection chain is exactly $[\tilde{h}_{i_0}^q]$. This is the sphere to which the homology lemma is applied.

9 Induction: eliminating handles below a chosen index

We now prove the first half of the normal form lemma.

Proposition 9.1. *Let $(W; M_0, M_1)$ be a compact smooth h -cobordism of dimension $n \geq 6$. Fix an integer q satisfying*

$$2 \leq q \leq n - 3.$$

Then W admits a handlebody decomposition relative to M_0 with no handles of index $< q$.

Proof. By Lemma 5.1, we may begin with a handlebody decomposition with no 0- or 1-handles. This proves the proposition for $q = 2$.

Now assume inductively that, for some r with

$$2 \leq r < q,$$

we have arranged that there are no handles of index $< r$. We will remove the r -handles, thereby obtaining a decomposition with no handles of index $< r + 1$.

Choose one r -handle, say $h_{i_0}^r$. Since there are no handles below index r , we have

$$C_{r-1}(\widetilde{W}, \widetilde{M}_0) = 0.$$

By the acyclicity of $C_*(\widetilde{W}, \widetilde{M}_0)$, the argument of Section 6 shows that

$$d_{r+1} : C_{r+1}(\widetilde{W}, \widetilde{M}_0) \longrightarrow C_r(\widetilde{W}, \widetilde{M}_0)$$

is surjective. Hence there exist coefficients $x_j \in \mathbb{Z}[\pi]$ such that

$$[\widetilde{h}_{i_0}^r] = \sum_{j=1}^{p_{r+1}} x_j d_{r+1}([\widetilde{h}_j^{r+1}]).$$

Start with a trivial framed embedding

$$\alpha : S^r \times D^{n-r-1} \hookrightarrow \partial_1 W_r.$$

Its core sphere has algebraic intersection chain

$$[\widetilde{\alpha}] = 0.$$

By Lemma 8.1, there is a framed embedding

$$\beta : S^r \times D^{n-r-1} \hookrightarrow \partial_1 W_r$$

whose core sphere is isotopic to the core of α inside $\partial_1 W_{r+1}$ and satisfies

$$[\widetilde{\beta}] = [\widetilde{\alpha}] + \sum_{j=1}^{p_{r+1}} x_j d_{r+1}([\widetilde{h}_j^{r+1}]) = [\widetilde{h}_{i_0}^r].$$

By Lemma 7.1, and using the Whitney trick, the core sphere of β is isotopic inside $\partial_1 W_r$ to an embedding that meets the belt sphere of $h_{i_0}^r$ transversely in exactly one point and misses the belt spheres of all other r -handles.

Thus β satisfies the two hypotheses of Lemma 4.1. It has the required single-intersection behavior before the $(r + 1)$ -handles are attached, and it becomes trivial after those handles are attached. Therefore Lemma 4.1 removes the chosen handle $h_{i_0}^r$ at the cost of adding one new $(r + 2)$ -handle.

Repeating this process finitely many times removes all r -handles. This completes the induction step. Hence, for the chosen index q , all handles of index $< q$ can be eliminated.

The important point is not that the total number of handles decreases. It need not. Rather, the lowest index present in the decomposition increases. In slogan form:

Below the chosen range, an r -handle can be pushed upward to an $(r + 2)$ -handle.

This proves the proposition. □

Example 9.2 (Running the low-index induction). Suppose $n = 8$ and we want the final normal form with $q = 3$. Lemma 5.1 first removes 0- and 1-handles. The induction then removes the remaining 2-handles; each removed 2-handle may create a new 4-handle. After this stage, there are no handles of index below 3. The new 4-handles are not a problem, because the later duality step is designed to remove handles above $q + 1 = 4$, while handles of index 4 are allowed to remain.

10 Dual handles

The preceding proposition eliminates low-index handles. To eliminate high-index handles, we use handle duality.

Suppose first that

$$W \cong M_0 \times [0, 1] \cup h^r$$

has a single r -handle relative to M_0 . If we reverse the cobordism and regard W as a cobordism from M_1 to M_0 , then the same geometric piece is a handle of index

$$n - r.$$

This is the dual handle. Geometrically, the core and cocore exchange roles:

$$D^r \times D^{n-r} \longleftrightarrow D^{n-r} \times D^r.$$

Thus an r -handle in the original decomposition becomes an $(n - r)$ -handle in the reversed decomposition.

For a full handlebody decomposition, the conclusion is that a decomposition relative to M_0 with p_r handles of index r gives a dual decomposition relative to M_1 with p_r handles of index $n - r$. Symbolically,

$$r \longleftrightarrow n - r.$$

Since $(W; M_0, M_1)$ is an h -cobordism, the reversed cobordism

$$(W; M_1, M_0)$$

is also an h -cobordism. Therefore the lower-index elimination argument applies equally well to the dual handle decomposition.

The key point is that high-index handles in the original decomposition become low-index handles in the reversed decomposition. Thus duality converts the problem of removing high-index handles into the problem already solved in Proposition 9.1.

Example 10.1 (Duality in dimension six). Return to the case $n = 6$. An original 4-handle becomes a dual 2-handle, and an original 5-handle becomes a dual 1-handle. Thus removing high-index handles in the original decomposition is exactly the same as removing low-index handles in the reversed decomposition.

11 Proof of the normal form lemma

We now prove Theorem 1.1.

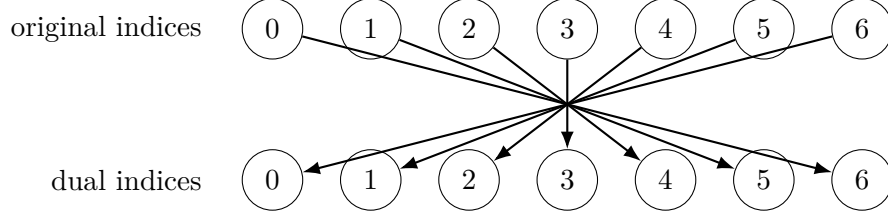


Figure 5: Duality of indices in dimension $n = 6$: an original r -handle becomes a dual $(6 - r)$ -handle. The same picture works in general dimension with $r \mapsto n - r$.

Proof. Fix q satisfying

$$2 \leq q \leq n - 3.$$

First apply Proposition 9.1 to the handlebody decomposition relative to M_0 . We obtain a decomposition with no handles of index $< q$. Thus all remaining handles have indices

$$q, q + 1, q + 2, \dots, n.$$

It remains to remove the handles of index $> q + 1$. To do this, pass to the dual decomposition relative to M_1 . In the dual decomposition, an original handle of index r becomes a handle of index $n - r$. Therefore original handles of index

$$r \geq q + 2$$

correspond exactly to dual handles of index

$$n - r \leq n - q - 2.$$

We now apply Proposition 9.1 to the reversed h -cobordism

$$(W; M_1, M_0)$$

with the integer

$$n - q - 1.$$

This is allowed because the hypothesis

$$2 \leq q \leq n - 3$$

is equivalent to

$$2 \leq n - q - 1 \leq n - 3.$$

The proposition eliminates all dual handles of index

$$< n - q - 1.$$

Equivalently, it eliminates all dual handles of index at most

$$n - q - 2.$$

Dualizing back, this removes all original handles of index at least

$$q + 2.$$

One small point is worth making explicit. Before dualizing, we had already removed original handles of index $< q$. In the dual decomposition, this means there are no dual handles of index $> n - q$. During the lower-index elimination in the dual decomposition, an s -handle is replaced by an $(s + 2)$ -handle. Since we only eliminate handles with

$$s < n - q - 1,$$

the newly created handles have index

$$s + 2 \leq n - q.$$

Thus the dual process does not create handles above $n - q$. After the dual elimination, the only possible dual indices are therefore

$$n - q - 1 \quad \text{and} \quad n - q.$$

Dualizing back, the only possible original indices are

$$q \quad \text{and} \quad q + 1.$$

Hence W admits a handlebody decomposition relative to M_0 of the form

$$W \cong M_0 \times [0, 1] \cup \bigcup_{i=1}^{p_q} h_i^q \cup \bigcup_{j=1}^{p_{q+1}} h_j^{q+1}.$$

This is the desired normal form. □

Once this normal form has been achieved, the relative chain complex has the form

$$0 \longrightarrow C_{q+1}(\widetilde{W}, \widetilde{M}_0) \xrightarrow{d_{q+1}} C_q(\widetilde{W}, \widetilde{M}_0) \longrightarrow 0.$$

Because (W, M_0) is acyclic, the map d_{q+1} is an isomorphism. After choosing lifts of the q - and $(q + 1)$ -handles, this differential is represented by a square matrix over $\mathbb{Z}[\pi]$. This matrix is the algebraic object used to define the Whitehead torsion of the h -cobordism.

Example 11.1 (The simply connected algebra). If $\pi = 1$, then $\mathbb{Z}[\pi] = \mathbb{Z}$. In the final normal form the differential is represented by an integral matrix

$$A : C_{q+1} \cong \mathbb{Z}^m \longrightarrow C_q \cong \mathbb{Z}^m.$$

Acyclicity says that A is an isomorphism of free abelian groups, so $\det(A) = \pm 1$. In this special case the algebra is ordinary integer linear algebra. For a general fundamental group, the same picture takes place over the noncommutative ring $\mathbb{Z}[\pi]$, which is why the normal form is the natural entry point to Whitehead torsion.

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